

Electronic Still Picture Camera Spatial Frequency Response Measurement

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Introduction

Electronic still picture cameras possess many advantages over conventional film cameras. Among these are real time access to the images, easy digital manipulation of the images, and precisely repeatable performance. The latter two characteristics, when combined with the inherent low noise characteristics of modern CCD chips, give electronic systems the potential to provide better quality images than film systems in terms of tone and color reproduction, and image noise. The one area in which fundamental barriers are still present in electronic systems is in high resolution image capture. Fine-grain films allow for the resolution of detail which is smaller than the smallest CCD pixel. In terms of total image resolution, it is also possible to manufacture films with areas many orders of magnitude greater than the largest CCD chips.

The accurate and meaningful measurement of resolution is therefore of prime importance to electronic imaging system development. While it is always possible for a film system to be designed which exceeds any electronic system in resolution, it is frequently possible to design electronic systems which provide adequate resolution, and beat film systems in other areas. The question is how to objectively define and measure resolution. This question is currently being addressed by the Resolution Subcommittee of ISO TC42/WG18.

In this committee, many issues have arisen. Some important ones are:

1. What are resolution, resolving power, limiting resolution, and visual resolution, and how do they relate to the MTF?
2. How do the resolution and noise characteristics of film and CCD chips vary, and how do these characteristics relate to perception?
3. How to measure the resolution characteristics of the components of an imaging system in a way which is interpretable and relates to the system as a whole.
4. How to relate pictorial electronic imaging system resolution measurements to film system measurements.

The obvious place to start in answering these questions is with film measurement standards. For pictorial photography, which is related to the general field of photography in much the same way as electronic photography is related to electronic imaging, two standards exist^{1,2}. Draft standards

also exist for optical transfer function measurement and application, and the measurement of the resolution of video camera systems^{3,4,5}.

The existence of this work both assists and complicates the development of the proposed electronic photography standard. Film is a random, isotropic imaging system but electronic systems are not. The goals in video imaging are subtly but significantly different than for pictorial still imaging. Data acquisition and noise characteristics are different in all the systems. While it is important that the new standard correlate with existing standards, it may not be desirable for the same techniques to be used.

In addressing the above issues over the past three years, progress has been made in several areas. The proposed standard will define three types of electronic photography resolution measurement:

1. Visual Resolution, which is similar to the current photographic resolving power;
2. Limiting Resolution, which is similar to the current video resolution; and
3. Spatial Frequency Response, which should give results which are comparable to the current photographic MTF, although significantly different techniques are used for determination.

While some issues are still to be worked out concerning visual and limiting resolution, the applicability of existing standards in these areas has reduced the amount of new research to be done. However none of the techniques used for the determination of the MTF are ideal for electronic photography. The photographic standard uses sine-wave targets and microdensitometer scans, and assumes an isotropic system. The optical standards could be applied, but do not deal with the limited sampling resolution, noise and data analysis issues specific to electronic photography systems. Also the two dimensional optical transfer function and related MTF possess considerably more information than photographers typically need, and are not easily comparable to photographic MTF's. The goal of the proposed standard is to outline reliable, repeatable techniques which will produce an electronic analog of the photographic MTF. This paper presents results of some of the efforts toward this goal.

Discussion

The first step in standardizing on an electronic photography MTF-type metric was to standardize on terminology. Ini-

tially, the metric was referred to as MTF, but as the techniques developed it became clear that the metric as defined in the proposed standard would be a very specific case of the MTF. It was felt that confusion might result if the general MTF term was used, and the *photographic* MTF was already specified in ANSI PH 2.39². Consequently, the metric was designated the Spatial Frequency Response (SFR). The SFR is the average of several MTF's obtained in specific ways. It represents the mean spatial frequency response of an electronic still picture camera.

The next step was to determine an approach. The photographic approach is partially valid — it produces average, one dimensional MTF curves, but assumes an isotropic medium and deals with aspects of microdensitometry that are not needed with electronic cameras. Also, the required use of sine-wave targets was felt by the committee to be inconvenient and probably unnecessary. Digital image processing techniques allow for the determination of MTF's from point and line spread functions in ways that are consistent with the optical transfer function standards. The challenge is to develop a technique which is efficient and easily applicable, but produces consistent and accurate results.

The current technique of choice is the determination of one-dimensional MTF's from a number of edge spread functions (ESF's). A test chart has been designed which contains elements for the determination of visual and limiting resolution, and also a number of black-to-white edges for the determination of these MTF's. The edges are placed at various positions on the chart and rotated at various angles to allow horizontal and vertical MTF's to be determined, and also the radial and tangential MTF's at one position. These MTF's are averaged to obtain the SFR of the camera. The illumination (and various other parameters of significance) used for these determinations are to be specified in the proposed standard.

This technique requires that a number of MTF's be determined for each SFR measurement, so it is imperative that an efficient and reliable technique for the determination of the individual MTF's be established. This paper will report on analyses that have been conducted to this end.

The Basic Algorithm

The technique for determination of MTF from the ESF was originally developed using a microdensitometer to scan a knife edge on film⁶. With CCD images, traditional knife edge techniques will yield inaccurate results due to the limited sampling resolution available. By using an extended knife edge technique⁷, an edge can be derived which is at a much higher resolution than the physical pixel spacing. This edge can then be used to determine the system MTF.

This edge is then processed to remove most of the noise in the data without degrading the ESF. The ESF is then differentiated to get a line spread function (LSF), which is then Fourier Transformed to get the MTF. Multiple MTF determinations are then combined to get an overall Spatial Frequency Response (SFR) for the camera. The following sections will describe the algorithms that are used to calculate the MTF from an edge.

Construction of the Super-Resolution Edge

The first step in determining MTF's from ESF's is the construction of a *super-resolution edge*. This solves the problem of the sampling frequency of the CCD chip being limited by the physical pixel spacing. In the past, this limitation has been avoided by scanning an edge across the detector; the "moving edge" technique. However the goal of the technique to be developed is for it to be applicable to a single image file — temporal data is not allowed. The proposed standard calls for the moving edge to be simulated in the stationary image through the use of a slanted edge⁷. Since only a one-dimensional edge profile is required for each measurement, tilting the edge slightly produces slight offsets in the position of the edge in a row as one moves from one scan line to the next.

The super-resolution edge is constructed by estimating the location of the edge for each scan line to sub pixel accuracy, using linear regression to fit a line to these estimates, and using this line to determine offset shifts to apply to each scan line for combination into the super-resolution edge. Linear regression on the edge location estimates results in a much more robust estimate of the actual edge locations, and reduces the influence of noise and the shape of the edge profile in determining the actual edge position.

Spacing of the Samples

One of the results of constructing the super-resolution edge in the manner just described is that the samples will not usually be evenly spaced. This can present a problem, since the differentiation and Fourier Transformation steps assume that the data is evenly spaced. In order to solve this problem, the data was collected into bins and averaged.

A difference between this work and that of Reichenbach⁷ was that in our work, the binning and averaging reduced the over-sampling resolution only to the extent necessary to obtain data that was (approximately) evenly spaced. Although averaging does reduce noise in the data, it only does so when a large number of samples are averaged, and can also cause excessive blurring of the edge. When looked at from a signal processing standpoint, averaging (sometimes called a *boxcar filter*) does not have very good pass-band characteristics (it blurs too much) or very good stop-band characteristics (it does not suppress noise as well as other filters). Subsequent sampling rate reduction was performed using filters with much better pass-band and stop-band characteristics than the boxcar filter, so as to more efficiently reduce the noise without biasing the resulting MTF calculations.

Decimation and Noise Reduction

After the binning of the edge data, we have the super-resolution edge represented by approximately evenly spaced samples. The binned super-resolution edge is still at a much higher sampling resolution than is necessary to accurately represent the data. By low-pass filtering the data and reducing the sampling rate (a process called decimation in signal processing⁸), we can greatly reduce the noise in the edge scan data while maintaining an accurate representation of the edge.

The decimation is performed by convolving the original function with a filter to remove frequency components

above the Nyquist limit, followed by the elimination of every other sample. The decimation step was performed in multiple passes, with each pass performing a decimation-by-two. By performing this as a series of decimation-by-two operations, better computational structures (both shorter and more efficient filters) can be used.

A particularly efficient class of filters can be used for decimation-by-two which are called *half-band* filters⁸. The half-band filter makes for an efficient implementation since nearly half of the terms in the convolution kernel are zero, and the convolution kernel is symmetric.

The decimation-by-two convolution is defined as follows:

$$E_h(x) = H(x) \bullet E(x) \quad [1]$$

where $\bullet$ denotes the convolution operator. The half-band filter used in this work had a 19-point convolution kernel⁹, which is shown in the following table:

Table 1. Half-band Decimation Filter Coefficients

$h(0)$	0.5
$h(1) = h(-1)$	0.307739
$h(2) = h(-2)$	0.0
$h(3) = h(-3)$	-0.077942
$h(4) = h(-4)$	0.0
$h(5) = h(-5)$	0.026184
$h(6) = h(-6)$	0.0
$h(7) = h(-7)$	-0.007080
$h(8) = h(-8)$	0.0
$h(9) = h(-9)$	0.001099

where

$$H(x) = [h(-9) \ h(-8) \ \dots \ h(8) \ h(9)] \quad [2]$$

At this point, the decimated edge data may still contain some noise. A further noise reduction step was performed using a median filter. The median filter will attenuate noise in uniform areas without blurring sharp transitions such as edges. The median filter is computed by taking a window of size p centered on a given sample, sorting the data in that window, and replacing the sample with the middle value (the median). This is done for each sample in the data.

$$E_f(x) = \text{Median}_p[E_{win}(x), p] \quad [3]$$

By using the median filter, we can significantly reduce the remaining noise in the data without blurring the edge. This is a very important property of the median filter, since if the edge data is blurred, the apparent MTF of the system will decrease.

Discrete Differentiation

Once the super-sampled edge has been decimated and low-pass filtered to reduce the noise, the line spread function (LSF) is computed by a discrete differentiation of the edge data. The earlier noise-reduction steps are very important, since the differentiation step will serve to greatly

amplify any high frequency noise remaining in the signal. Discrete differentiation is performed by convolving a differentiating filter $D(x)$ with the edge data:

$$L(x) = D(x) \bullet E_f(x) \quad [4]$$

A simple differentiating filter, which was used in the examples shown later, is:

$$D(x) = [-1 \ 0 \ 1] \quad [5]$$

Windowing and Transformation into the Frequency Domain

The final step in determining the SFR is to take the Fourier Transform of the LSF, using the Fast Fourier Transform (FFT) algorithm. Before taking the FFT, though, the data needs to be windowed. The unwindowed LSF is, in reality, a continuous function that has been implicitly windowed by a rectangular window whose width is that of the data. It is well known in signal processing literature^{10,11} that a rectangular window in the time domain contains a great deal of high frequency energy. When the FFT is performed, some of this energy will show up in the lower frequency components. This leakage of energy distorts the resulting frequency response.

In order to minimize this leakage, the data is windowed with a function which gradually rises from zero to a maximum and back to zero over the width of the data. There have been many windows proposed in the literature^{10,11}. For this work, the Welch window was used:

$$W(x) = 1 - \left(\frac{x - \frac{1}{2}(N-1)}{\frac{1}{2}(N+1)} \right)^2 \quad [6]$$

where N is the length of data to be windowed, and x is the sample number ($0 \leq x \leq N-1$). The Welch window was chosen since it provides good leakage reduction, but does not significantly distort the low frequency response¹⁰. Note that when windowing the LSF, a window that falls to zero too rapidly will narrow the LSF, which will result in artificially *increasing* the apparent MTF of the system.

As a practical matter, the LSF data $L(x)$ are centered under the window, so the peak of the LSF and the peak of the window coincide. The centered LSF data is then multiplied on a sample by sample basis by the window function:

$$L_{win}(x) = L(x) \cdot W(x) \quad [7]$$

Finally, the windowed LSF data is zero padded at the ends to reach a data length that is a power of two, and is Fourier Transformed using the FFT. The padding of the data serves two purposes. First, very efficient FFT algorithms are available for a data length that is a power of two. Second, by increasing the sequence length of the data to be transformed, the spectral resolution of the transform is improved. Since we know that the LSF goes to zero at the ends of the data range (due to both the nature of the LSF and the windowing step), the zero padding will not introduce any biases in the final result.

Experimental Results

For experimental verification, the algorithm as described above was applied to image files from a digital still picture camera. The results from two specific image files will be shown, which were taken at the exposure index setting limits of an electronic still camera. The low setting results in noise at a level which is not generally visible for typical photographic images, and are therefore probably at the lower limit of electronic photography system design. The high setting results in noise at a level which is becoming visually objectionable, and is therefore close to the upper design limit. The purpose of this study was to determine an appropriate algorithm for MTF and SFR determination. The curves presented below, while representative of a camera's system performance, are not intended to be final SFR determinations.

For these experiments, a 64x64 pixel window of a slanted edge was taken. Figure 1 shows the super-resolution edge which was created from an image captured at the low setting on the camera. This plot shows the data before any binning or decimation steps were performed.

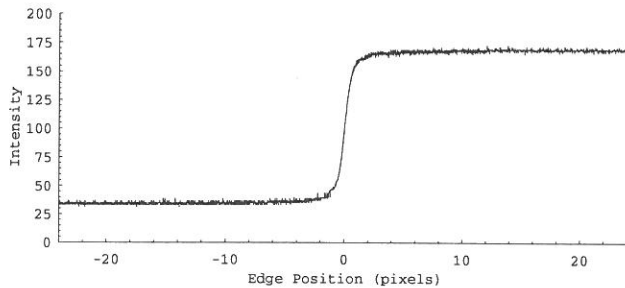


Figure 1. Super-resolution edge (Low setting)

The data was then binned to obtain a 14x oversampled super-resolution edge (the slope of the edge was such that it shifted approximately one pixel every 14 scan lines). Decimation by a factor of 4 was performed to obtain smoothed data at a sampling rate of 3.5x. This rate is high enough to retain complete edge information. A median filter with a length of 15 samples was then applied to smooth the remaining noise. The data was windowed and Fourier Transformed to get the SFR.

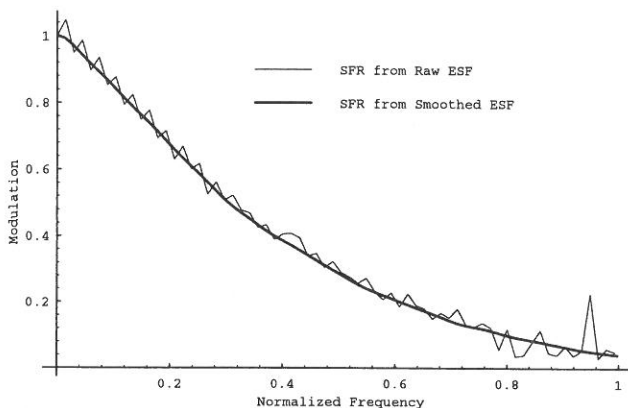


Figure 2. Spatial Frequency Response (Low setting)

Figure 2 shows a plot of the SFR that was obtained (plotted against a normalized frequency). The thick line shows the results from applying the algorithms described in this paper. The thin line shows the result of computing the SFR by differentiating and Fourier Transforming the raw super-resolution edge. The differences between the two sets of data are a result of the noise in the original data.

In order to test the robustness of the algorithm, an image containing a significant amount of noise was also analyzed. Figure 3 shows the super-resolution edge which was created from an image captured at the high setting on the camera. This plot again shows the data before any binning or decimation steps were performed.

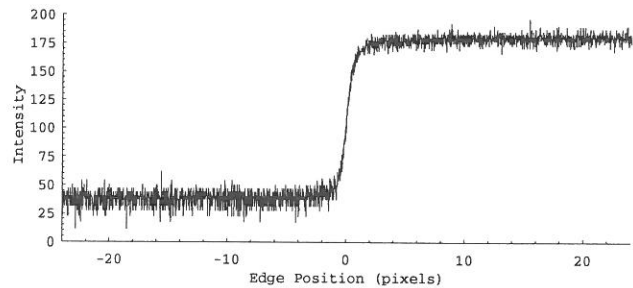


Figure 3. Super-resolution edge (High setting)

This data was processed in exactly the same manner as the low setting edge. Figure 4 shows a plot of the SFR that was obtained. Again, the thick line shows the results from applying the algorithms described in this paper, while the thin line shows the result of computing the SFR by differentiating and Fourier Transforming the raw super-resolution edge. In this case, the noise makes it difficult to draw any meaningful conclusions from the unfiltered response curve.

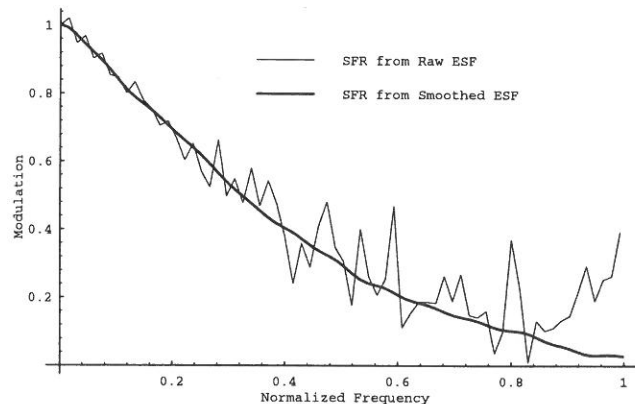


Figure 4: Spatial Frequency Response (High setting)

The individual SFR's obtained from the low and high settings on the camera were also compared. The results obtained using the algorithm described in this paper matched extremely well over the full frequency range, demonstrating the robustness of the algorithm to different levels of noise. The final SFR reported for a camera will actually be an av-

erage of a number of these individual SFR's, measured at different locations in the camera's field of view.

Conclusions

An important characteristic of electronic still picture cameras is their image resolution. The Resolution Subcommittee of ISO TC42/WG18 has been working on standardizing the methodology for measuring and reporting resolution for these cameras, to allow users to compare different camera systems.

In this paper, an algorithm has been presented for determining Spatial Frequency Response (SFR) for an electronic still picture camera which does not require an elaborate test setup or expensive test equipment. This algorithm determines the SFR by electronically processing test image data captured with the camera. It can accurately determine the SFR under widely varying noise conditions, is computationally efficient, and simple to implement.

Acknowledgments

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Advantages of Tag-Based Formats for Electronic Still Photography

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Overview of Current Standards Work on "Removable Memory"

The ISO TC42/WG18 group is chartered to develop international standards for electronic still picture imaging. The NAPM IT10 group is similarly chartered to develop US national standards for electronic still picture imaging and to provide US input to the ISO group. Currently, a work item entitled "Photography - Electronic Still Picture Cameras - Removable Memory" is underway in both of these groups. The goal is to develop an image storage standard for electronic still cameras.

The OSI/ISO Basic Reference Model¹ is being used to guide the Removable Memory work. This model has seven

layers, as shown in Table 1, ranging from the lowest "physical" layer to the highest "application" layer. The two upper layers, application and presentation, i.e., the content and structure, comprise the Image Data Format. The image data features of the Image Data Format form the application layer. The encoding mechanism used to format the image data features the presentation layer. The layers below the application and presentation layers are classified as the Image Storage Application Profile or ISAP.

The ISO and NAPM groups have recognized that different types of media will be used to store images in electronic still cameras. Manufacturers will select the appropriate removable memory depending on the application and end-user requirements. Removable memory types include solid-